

Business Forecasting: Project 2 (Programming & Presentation)

In today's competitive retail landscape, accurate sales forecasting is essential to achieving low operational costs while maintaining high customer satisfaction. For businesses with hundreds of store locations and diverse market conditions, even small improvements in forecast accuracy can translate into significant financial and operational benefits.

Your client is a large European drugstore chain with hundreds of retail locations operating across diverse regions. The business operates on a daily sales cycle, and performance is influenced by multiple dynamic factors such as promotional campaigns, holiday schedules, store-specific characteristics, and regional competition.

This project was initiated under the sponsorship of the company's Chief Operations Officer and Lead Data Scientist, who sought your external forecasting expertise to evaluate and enhance their current forecasting capabilities. As a team of data science consultants, you are tasked to design a robust forecasting approach tailored to the complexities of their multi-store retail environment.

Your task is to analyze historical performance, uncover patterns and drivers of sales variability, and develop a forecasting framework that is both accurate and operationally practical. This involves understanding how factors such as store operations, promotions and holidays impact weekly sales and building models that can incorporate these signals effectively.

The insights and solutions developed in this project will inform tactical decisions (e.g. weekly inventory and staffing plans) as well as short-term financial planning (e.g. budgeting, short-term cashflow), providing a valuable foundation for data-driven decision-making at scale. Therefore, the company decided **as output** on **weekly sales forecasts for the next 6 weeks**. While the COO is extremely interested in **accurate weekly forecasts for the next two weeks** for production and logistic planning, the CFO of the company is rather interested in the **sum of sales over the next 6 weeks** to steer the cashflow.

After providing the code and forecasting framework, you should also present your findings in a poster session to the companies COO and data science team to inform them about your results and the suggested forecasting framework.

Dataset

The file `sales_data.csv` contains the following data points on a daily basis:

- `store_id`: a unique identification for each store
- `sales`: the turnover for any given day
- `customers`: the number of customers on a given day
- `open`: an indicator for whether the store was open: 0 = closed, 1 = open
- `state_holiday`: indicates a state holiday. Normally all stores, with few exceptions, are closed on state holidays. Note that all schools are closed on public holidays and weekends. a = public holiday, b = Easter holiday, c = Christmas, 0 = None
- `school_holiday`: indicates if the store_included was affected by the closure of public schools on that date
- `promo`: indicates whether a store is running a promo on that day

The file `future_values.csv` contains future values of some of the features in the sales data to be used for your out-of-sample forecast if you decide to go with a multivariate approach.

The file `metadata.csv` contains metadata for each store with the following fields:

- store_id: a unique identification for each store
- store_type: differentiates between 4 different store models: a, b, c, d
- assortment: describes an assortment level: a = basic, b = extra, c = extended
- competition_distance: distance in meters to the nearest competitor store

Task 1: Programming

The task is much more simply described than executed: Create a python script that creates the weekly forecast for all stores, the total for each store type and the overall total for the company for the **upcoming 6 weeks**. The company is looking for accurate and reasonable forecasts while maintaining a manageable effort in the forecasting process.

Nonetheless, the complexity of the task is high as you could consider for example:

- As in project 1, a one-size-fits-all approach is unlikely to deliver the best results. Think about partitioning your data or giving enough flexibility within your approach to model different patterns / consumer behaviours.
- Using univariate statistical methods, multivariate machine learning methods or even deploy a global model.
- Complexity vs. maintainability and run time of your approach.
- Explainability of your approach.

While at the end the company wants you to deliver one final approach being implemented, it might make sense to try several approaches (e.g. benchmarks) to compare the accuracy and use this for reasoning in Task 2. In order to understand your approach better, the client's data science team has asked for a comprehensive list of your modelling decisions.

Remarks

- **All additional assumptions, decisions and most importantly the reasoning behind them** should be stated in an `.pdf`.
- You can assume that the input data / files always have the same format and are stored in the same directory as your Python file.
- You should only use forecasting methods discussed in the lecture and exercises. If you want to use any method not discussed in the lecture, you must be able to explain it during the poster session.
- You should mostly use the Python libraries provided in the extended environment at the beginning of the lecture. If you need additional libraries, they must be available in the Python Package Index (pypi.org). Explain in a comment, why you need them.
- Your code should be structured and commented to a reasonable extent.

Task 2: Presentation during poster sessions

After creating your forecasting framework, you should present it to the supposed COO and data science team in a poster session on Tuesday, July 14th at 2:15-5:45 p.m. Templates for the posters are provided.

Your poster should at least include (not necessarily in this order):

- Shortly the main results of your data analysis and necessary data preparation.
- A description of the forecasting framework.
- An approximation of the achieved accuracy with focus on business relevant measurements.
- A (business) reasoning why you suggest this particular forecasting process.
- A short plan how often it needs to be updated / retrained / maintained.

We will print the posters for you. During the poster session, there will be a mandatory Q&A to your findings.

Remarks

- The poster is for a COO and the lead data Scientist. So you have to strike a balance between overall simplified information and some deeper analysis and technological details.
- Feel free to support your analysis with pictures and plots, but make sure they are relevant for your findings.

Poster requirements

The poster must fulfill the following requirements:

- Size: A0
- Minimal resolution is 300 dpi (this can be ensured in the settings when converting to pdf).
- To ensure there is nothing cut off by the printing team, please leave 3mm from each edge.

Submission

The submission must include:

- One single .py file with necessary comments (see Remarks) included.
- If necessary, any additional files for the script to run (e.g. saved models, additional imports).
- One single .pdf file stating all your assumptions and decision made for modelling. Please include the AI declaration at the end.
- One single .pdf file with the poster.

Additionally, there are the following rules:

- The full project has to be submitted in the Moodle course under section `Project 2` latest until 12.07.2025 at 11:30 p.m.
- All submissions must follow the chair-specific AI guidelines (see Moodle) and TUM regulation. In particular, you need to include a 'Declaration of AI Usage'. Please add it for both parts at the end of your report.
- Every student has to submit all files individually. For registered groups, files can be the same.
- For your submission to be graded, you must be registered in tumonline for the exam.

Groups & Group registration

You are allowed to work in groups of up to 2 people. Both group members must be present at the poster session and answer questions in the Q&A.

If you decide to work in groups, please send a mail to dominik.proschmann@tum.de with your partner in cc. Remember that only communication using your official @tum.de email address is allowed.